Extra Bite #15: Modulators, Mixers, and Detectors

Reactance and balanced modulators, FM generation, mixer products, envelope and product detectors.

Group E7E · 11 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

Generating FM (Reactance Modulator)

To modulate an oscillator's frequency, you change the reactance of its tank circuit in step with the audio. A **reactance modulator** fakes a variable capacitance (or inductance) in parallel with the tank.

Question	Answer	Q
How to generate FM phone signals	**Reactance modulation of a local oscillator** (E7E01). Vary the tank reactance; the oscillator frequency follows.	E7E01
What does a reactance modulator do?	**Produce PM or FM signals by varying a capacitance** (E7E02). Typically a varactor driven by the audio.	E7E02

Generating SSB (Balanced Modulator + Filter)

A **balanced modulator** multiplies the audio by the carrier and cancels the carrier itself, leaving both sidebands (DSB-SC). A sharp filter then keeps just one sideband.

One way to produce SSB: use a balanced modulator followed by a filter (E7E04). Balanced mod cancels the carrier; filter removes the unwanted sideband.

FM Pre-emphasis and De-emphasis

FM rolls off naturally at high audio frequencies relative to noise. To compensate, transmitters boost the highs before modulation; receivers cut them back after demodulation.

Question	Answer	Q
What boosts higher audio frequencies in an FM speech channel?	**A pre-emphasis network** (E7E05). Added at the transmitter, before modulation.	E7E05
Why de-emphasis at the receiver?	**For compatibility with transmitters using phase modulation** (E7E06). Many "FM" rigs actually use PM with pre-emphasis; receiver de-emphasis flattens the audio back out.	E7E06

Demodulators

Detector	What it does	Q
Frequency discriminator	**A circuit for detecting FM signals** (E7E03). Converts instantaneous frequency back into audio voltage.	E7E03
Diode envelope detector	Works **by rectification and filtering of RF signals** (E7E10). Standard AM detector: diode rectifies, RC filter smooths.	E7E10
Product detector	**Used for demodulating SSB signals** (E7E11). Multiplies the incoming SSB signal by a beat-frequency oscillator at the suppressed-carrier frequency, recovering the audio.	E7E11

Mixers And Baseband

A mixer takes two input signals and outputs the sum and difference frequencies (and the original two). It is the fundamental frequency-shifting building block of every superhet.

Question	Answer	Q
Principal frequencies at a mixer output	**The two input frequencies along with their sum and difference frequencies** (E7E08).	E7E08
What if mixer input levels are too high?	**Spurious mixer products are generated** (E7E09). Driving the mixer past its linear range creates extra unwanted tones.	E7E09
What is "baseband"?	**The frequency range occupied by a message signal prior to modulation** (E7E07). For voice, baseband is the audio band; for digital, it is the bit-rate spectrum.	E7E07

Mental Cheat Sheet

- FM generated by **reactance modulation of a local oscillator**.
- Reactance modulator varies a **capacitance** to make PM or FM.
- SSB by **filter method**: balanced modulator + filter.
- FM transmit: **pre-emphasis** boosts highs before modulation.
- FM receive: **de-emphasis** for compatibility with PM transmitters.
- Discriminator detects **FM**.
- Envelope detector = diode + RC, detects **AM**.

Common Traps

- Reactance modulator works by varying **capacitance**, not resistance.
- SSB generation uses **balanced modulator + filter**, not "reactance + mixer."
- De-emphasis exists for **PM-transmitter compatibility**, not noise reduction or efficiency.
- Envelope detector works by **rectification and filtering**, not by Zener breakdown.

- Product detector = multiplier with BFO, detects **SSB**.
- Mixer outputs: both inputs + sum + difference.
- Overdriving a mixer creates **spurious products**.
- Baseband = the message signal before modulation.

- SSB needs a **product detector**, not a discriminator or phase detector.
- Mixer output has both inputs PLUS their sum AND difference; trap answers say “2× and 4×” or “square-root products.”
- Overdriven mixer → **spurious products**, not “automatic limiting.”
- Baseband is **pre-modulation**, not the basic oscillator frequency.

Takeaway: Modulation puts the message on the carrier; demodulation takes it back off. FM is generated by reactance modulation and recovered by a discriminator. SSB is generated by a balanced modulator plus a filter and recovered by a product detector. AM is detected by a diode envelope detector. Mixers produce sum-and-difference frequencies and create spurious products when overdriven. Baseband is the message before modulation.

Bite 15 Question Bank (E7E)

E7E01. Which of the following can be used to generate FM phone signals?
A. Balanced modulation of the audio amplifier.
B. Reactance modulation of a local oscillator.
C. Reactance modulation of the final amplifier.
D. Balanced modulation of a local oscillator.

E7E02. What is the function of a reactance modulator?
A. Produce PM or FM signals by varying a resistance.
B. Produce AM signals by varying an inductance.
C. Produce AM signals by varying a resistance.
D. Produce PM or FM signals by varying a capacitance.

E7E03. What is a frequency discriminator?
A. An FM generator circuit.
B. A circuit for filtering closely adjacent signals.
C. An automatic band-switching circuit.
D. A circuit for detecting FM signals.

E7E04. What is one way to produce a single-sideband phone signal?
A. Use a balanced modulator followed by a filter.
B. Use a reactance modulator followed by a mixer.
C. Use a loop modulator followed by a mixer.
D. Use a product detector with a DSB signal.

E7E05. What is added to an FM speech channel to boost the higher audio frequencies?
A. A de-emphasis network.
B. A harmonic enhancer.
C. A heterodyne enhancer.
D. A pre-emphasis network.

E7E06. Why is de-emphasis used in FM communications receivers?
A. For compatibility with transmitters using phase modulation.
B. To reduce impulse noise reception.
C. For higher efficiency.
D. To remove third-order distortion products.

E7E07. What is meant by the term “baseband” in radio communications?
A. The lowest frequency band that the transmitter or receiver covers.
B. The frequency range occupied by a message signal prior to modulation.
C. The unmodulated bandwidth of the transmitted signal.
D. The basic oscillator frequency in an FM transmitter that is multiplied to increase the deviation and carrier frequency.

E7E08. What are the principal frequencies that appear at the output of a mixer?
A. Two and four times the input frequency.
B. The square root of the product of input frequencies.
C. The two input frequencies along with their sum and difference frequencies.
D. 1.414 and 0.707 times the input frequency.

E7E09. What occurs when the input signal levels to a mixer are too high?
A. Spurious mixer products are generated.
B. Mixer blanking occurs.
C. Automatic limiting occurs.
D. Excessive AGC voltage levels are generated.

E7E10. How does a diode envelope detector function?
A. By rectification and filtering of RF signals.
B. By breakdown of the Zener voltage.
C. By mixing signals with noise in the transition region of the diode.
D. By sensing the change of reactance in the diode with respect to frequency.

E7E11. Which type of detector is used for demodulating SSB signals?
A. Discriminator.
B. Phase detector.
C. Product detector.
D. Phase comparator.

Answer Key

E7E01: B	E7E02: D	E7E03: D	E7E04: A
E7E05: D	E7E06: A	E7E07: B	E7E08: C
E7E09: A	E7E10: A	E7E11: C	